

CAEAID: An incremental contrast learning-based intrusion detection framework for IoT networks

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ABSTRACT

Nowadays, the swiftly advancing and intricately diverse IoT node devices produces high-dimensional, discrete, and temporally dynamic network traffic feature data. The ensuing data distribution sparsity and concept drift can critically impair the effectiveness of traditional deep learning-based intrusion detection models. To address these issues, we propose an incremental contrastive learning-based intrusion detection framework for IoT networks, CAEAID. On one hand, to tackle the high-dimensional sparse distribution of traffic, we construct a contrastive autoencoder. It effectively learns low-dimensional latent representations of IoT traffic features by minimizing the distance between similar samples while maximizing the distance between dissimilar samples. Subsequently, we identify abnormal traffic based on distance. The contrastive autoencoder clarifies the boundaries of traffic categories and alleviates the challenges posed by high-dimensional sparse spaces. Simultaneously, we apply improved extreme value theory to fit IoT traffic features and adaptively establish thresholds for detecting extreme discrete anomalous traffic for auxiliary analysis. On the other hand, to handle concept drift, CAEAID creates a pseudo-labeled dataset based on detection consistency, enabling incremental learning and periodic model updates for adaptive detection. Experimental results indicate that compared to other advanced methods, CAEAID improves the accuracy on the IoTID20 and CICIDS2018 datasets by at least 1.15% and 1.72%, respectively. Furthermore, the framework demonstrates superior performance in precision, recall, and F1-score.

1. Introduction

In the era of the Internet of Everything, Internet of Things (IoT) technology is transforming our lives and work at an astonishing speed. This extensive connectivity among people, machines, and objects has fostered advancements across various domains, including industrial automation [1], smart transportation [2], intelligent cities [3], efficient healthcare [4], and smart homes [5]. It significantly enhancing societal production and living standards. However, the number of IoT devices is vast, and most of them have low-security performance, which leaves them vulnerable to attacks. The IoT networks involve diverse heterogeneous network environments that can be easily exploited by malicious actors. The increasing number of cyberattacks may threaten the confidentiality, integrity, and availability of IoT devices and systems.

There is an urgent need for effective intrusion detection solutions specifically designed for the IoT environment to protect IoT devices from potential attacks [6]. These solutions can swiftly detect security threats and intrusions, thereby enhancing the overall security of IoT

devices and systems. In recent decades, the exponential growth of IoT network data, coupled with the data-driven nature of artificial intelligence technology, has stimulated extensive research [7–9] into machine learning (ML)-based intrusion detection techniques. However, applying machine learning to intrusion detection research still faces many challenges. In a highly insightful paper [10] presented at the 2010 S&P Network Security Conference, Sommer and Paxson highlighted that the distinctive characteristics of the intrusion detection field pose greater challenges for the effective application of machine learning methods compared to many other scenarios.

Recently, more researchers have increasingly focused on effectively combining advanced cutting-edge deep learning (DL) technologies with intrusion detection in IoT scenarios. Numerous valuable research works [11,12] have been published. They lead to significant breakthroughs in the automatic feature extraction of IoT communication traffic, massive data processing. Despite these advancements, ongoing research has revealed that some challenges identified by Sommer and Paxson continue to persist. The heterogeneous, open,

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and constantly evolving nature of the IoT network ecosystem presents additional complexities. This paper aims to address the distribution sparsity and concept drift issues [13] that arise from the dynamic characteristics of IoT networks and the evolution of attacks.

Specifically, current research on the analysis of IoT traffic data and anomaly detection typically presumes that traffic data operates within a static or fixed environment characterized by a stable and tight distribution [14]. Consequently, the stable features observed in the training data are deemed sufficient to represent the traffic to be evaluated. However, the IoT environment is inherently dynamic and open. Its traffic data is scattered and sparse. While attackers continuously adapt their methods in response to evolving defense mechanisms. These factors contribute to temporal changes in the distribution and characteristics of traffic data within the IoT networks. In such scenarios, a static ML or DL model may fail to identify new attack traffic. The challenges mentioned above be summarized as the sparse distribution and concept drift issue of traffic data in IoT network intrusion detection research.

In the realm of intrusion detection in IoT networks, the challenges posed by sparse distribution and concept drift of traffic data can generally be addressed through two primary strategies: active drift detection and passive adaptation mitigation methods. Active drift detection [15–19] specifically targets the issue of concept drift. It involves monitoring the stability of IoT network data distributions or real-time performance metrics of models to detect and further adapt to concept drift. In response to sparse data distribution and concept drift, researchers have developed various passive adaptation and mitigation methods [20–22]. They primarily adjust the IoT traffic data used for training and the detection model continuously, enabling the model to incrementally learn new data distributions based on representative data.

Despite numerous proposed solutions for detecting and adapting to discrete data distribution and concept drift in IoT intrusion detection, existing methods still have limitations. Drift detection involves setting drift thresholds, which significantly impacts detection outcomes. Passive adaptation and mitigation strategies encompass two key aspects. Firstly, they relate to the obsolescence of historical traffic data. In an open network environment, the recurrence of historical traffic patterns can hinder the model's capacity to differentiate between normal and attack traffic. Secondly, these strategies involve the establishment of foundational models for learning paradigms, such as contrastive learning. If the foundational model is inadequately configured, it may introduce biases or overfitting issues, resulting in the inaccurate processing of new data and, ultimately, diminishing the effectiveness of incremental learning.

To solve above problems, we design an incremental contrast learning-based intrusion detection framework for IoT networks, CAEAID, that incrementally learns, analyzes, and detects from continuously evolving network traffic data. The framework constructs a contrastive autoencoder (CAE) to learn from both normal and attack traffic, it effectively captures low-dimensional representations of traffic features. It clarifies the boundaries between different traffic categories and alleviates the “dimensional curse” effect of high-dimensional sparse data on detection functions. Furthermore, CAEAID employs an improved extreme theory to conduct modeling analysis and adaptive threshold setting based on each traffic characteristics. It enables the detection of abnormal traffic samples, focusing on the tail information of the characteristic data, and assists in analyzing the detection results generated by the contrastive autoencoder model. It improves the stability of the underlying model. Subsequently, a consistency analysis is performed based on the detection results of both methods, leading to the generation of datasets with pseudo-labels to augment the initial training data. This process facilitates data augmentation and model incremental learning, enabling the detection framework to achieve adaptive intrusion detection (AID) for IoT network traffic and counter concept drift. During the inference stage, the contrastive autoencoder produces a low-dimensional latent representation of the

test samples. The CAEAID framework introduces a robust distance function to rapidly generate detection results and enhance detection efficiency. In summary, our primary contributions are outlined as follows:

1. We introduce the CAEAID framework to identify attack behaviors in IoT traffic. Specifically, we integrate a contrastive autoencoder model with improved extreme value theory, employ incremental learning and model updating to effectively detect intrusion behaviors and mitigate the effects of data sparsity and concept drift on intrusion detection systems in IoT networks.
2. CAEAID develops an intrusion detection model based on contrastive autoencoder to derive effective low-dimensional representations of IoT traffic data. It ensures intra-class compactness, inter-class separability, well-defined category boundaries, while also constructing a robust distance function to enhance detection performance and alleviate the negative impact of high-dimensional discrete traffic data on detection function.
3. CAEAID designs an anomaly detection of traffic characteristics based on improved extreme value theory for flow feature modeling. It adaptively establishes anomaly thresholds to detect extreme discrete values in IoT network traffic effectively, while also offering supplementary analysis of the detection results from the contrastive autoencoder model.
4. CAEAID performs a consistency analysis of the detection results obtained from contrastive autoencoders and improved extreme value theory. It is capable of generating pseudo-labels for certain testing traffic data, which can be utilized for training data augmentation and incremental learning of the detection model. It facilitates periodic updates of the detection framework to progressively adapt to concept drift.
5. Experiments utilizing the IoTID20 [23] and CICIDS2018 [24] datasets demonstrate that CAEAID has achieved significant improvements over previous state-of-the-art methods, with accuracy increases exceeding 1.15% and 1.72%, respectively, and F1-scores improved by more than 0.74% and 1.90%, respectively.

The rest of this paper includes the related work in Section 2, the CAEAID framework in Section 3, the experimental evaluation and analysis in Section 4, and the conclusion in Section 5.

2. Related work

This section covers the work related to the research in this paper. We explore two areas of related work: ML and DL-based IoT intrusion detection and major solutions for sparsity and concept drift of traffic data.

2.1. Researches on ML and DL-based IoT network intrusion detection

In recent years, ML and DL technologies have been widely applied in IoT intrusion detection research due to their automated learning, excellent data analysis, and inference capabilities, which aim to enhance detection accuracy and efficiency.

In the application of machine learning, Khan et al. [25] utilized an AutoML framework to select the optimal detection model. They applied the soft voting method to optimize the ensemble strategy, aiming to enhance the detection rate of attack classes in the UNSW-NB15 and CICIDS2017 datasets while minimizing false positives. Ye et al. [26] introduced a feature selection method based on hybrid breeding optimization to construct an optimal feature subset for intrusion detection samples, aiming to improve the intrusion detection accuracy of ML classifiers such as Support Vector Machine and Extreme Gradient Boosting (XGBoost).

In the application of deep learning, Gu et al. [27] optimized a Wasserstein Generative Adversarial Network using a self-attention

mechanism to generate new traffic based on existing anomalous traffic, which was used to train an improved Deep Neural Network (DNN) for enhancing the accuracy of multi-class anomaly traffic detection of Industrial IoT (IIoT). Jin et al. [28] proposed an IDS called FI-IDS for heterogeneous networks such as IoT networks, which utilized federated learning and provided a distributed architecture for IoT intrusion detection, achieving a balance between privacy protection and detection performance and enhancing detection robustness. Wanjau et al. [29] combined Convolutional Neural Network (CNN) and Bi-directional Long Short-Term Memory to learn spatial and temporal features of network traffic data comprehensively. Latif et al. [30] transformed network data into image data for practical analysis by CNN. They then used a genetic algorithm to fine-tune the hyperparameters of the learning model. They provided a solution for developing adaptive IDS in complex IIoT environments. Mao et al. [31] proposed a hierarchical federated incremental learning method for IIoT intrusion detection. It involved distributed model training and continuous learning on multiple devices, balancing intrusion detection and device operation constraints using a weighted transmission optimization strategy based on edge clients.

In the hybrid application of ML and DL algorithms, Vo et al. [32] have developed a rule-based approach and a ML model called PAID for hierarchical traffic inspection. PAID integrates DNN, XGBoost, and Gradient Boosting Machine to enhance the efficiency and performance of traffic data analysis.

Although the aforementioned methods have made significant progress in IoT intrusion detection, these studies generally rely on the assumption that traffic data remains static. The heterogeneous and open nature of IoT can lead to changes in data distribution, thereby altering traffic feature patterns. In such scenarios, models trained on static data may face the risk of significant degradation in detection performance.

2.2. Researches on sparsity and concept drift of traffic data in IoT network intrusion detection

To solve the problem of poor model generalization ability caused by sparse data and concept drift in IoT network intrusion detection, researchers have proposed various coping techniques. We divide them into active drift detection and passive mitigation adaptation.

In the aspect of active drift detection, Liu et al. [33] measured local drift by continuously monitoring the variances and means of the flow data distribution, effectively assessing whether there is concept drift between network data samples. Yang et al. [34] designed an IoT anomaly detection framework called PWPAAE, combining two drift detection methods, Adaptive Window and Drift Detection Method, along with various improved ML models to construct multiple base learners. PWPAAE achieved IoT anomaly detection by weighted averaging the performance of multiple anomaly-based detection models. Priya et al. [35] proposed CIDD-ADODNN, a deep learning model based on the Adadelta optimizer. This model used adaptive sliding window drift detection technology and designed a novel optimizer to enhance detection performance. Ultimately, the model achieved a detection accuracy of 95.92% on the KDDCup99 traffic dataset. Mahadi et al. [36] combined the disagreement measure with the Page-Hinkley test for drift detection in the context of traffic data. Yang et al. [37] introduced the CADE system and a distance-based drift explanation method. This system can effectively detect drift samples in intrusion detection in two networks and provide semantically meaningful explanations.

Regarding passive mitigation adaptation, Mulimani et al. [38] proposed an intrusion detection method based on Adaptive XGBoost technology to detect attacks in real-time environments such as the IoT and alleviate the problem of poor generalization ability caused by the concept drift and the dynamic change of network traffic. Wang et al. [39] designed an ensemble system named ENIDrift for intrusion detection in real networks like the IoT network. It integrated multiple AutoEncoders (AE) to generate new sub-classifiers by measuring the

stability of data chunks and continuously adjusting their weights to achieve model updates. Jain et al. [40] combined concept drift analysis based on sliding windows with k-means clustering to reduce the traffic data size used for training and update the traffic training set. They then retrained the detection model to mitigate the impact of concept drift on the detection model and improve intrusion detection accuracy. Sun et al. [41] constructed multiple base classifiers for stacked ensemble to address concept drift in industrial IoT intrusion detection and improve detection performance and the ability to detect unknown attacks through multiple retraining iterations. Andresini et al. [42] proposed INSOMNIA. This semi-supervised intrusion detection framework combined DNN and active learning to effectively enhance the model's ability to detect intrusions when traffic distribution changes.

The above research advances IoT intrusion detection in data sparsity and concept drift but faces limitations. Most drift detection methods rely on threshold settings, where inaccuracies compromise detection performance. For sparse traffic features, building stable models to obtain compact low-dimensional representations remains critical. Drift mitigation requires incremental learning and model updating, but effective new data integration remains challenging. We propose the CAEAID framework, which introduces an intrusion detection model based on contrastive autoencoder to learn low-dimensional deep representations of traffic feature stably. We construct an improved extreme value theory to adaptively fit and detect anomalies in IoT traffic feature data sequences, assisting in the analysis of the detection results from the contrastive autoencoder. By supplementing some testing data into the training data based on the consistency of the detection results, training data augmentation and periodic model updates are achieved to enhance the performance of IoT intrusion detection and improve its generalization ability.

3. CAEAID framework

3.1. Research motivation and problem statement

IoT intrusion detection faces two critical challenges: sparse traffic data distribution and concept drift. Device heterogeneity, diverse communication modes, and low-frequency attacks contribute to sparse traffic features and patterns. Meanwhile, environmental dynamics (e.g., network changes) and evolving attack strategies drive continuous shifts in traffic behavior. Both severely challenging the performance of traditional Network Intrusion Detection Systems (NIDS).

Taking the IoTID20 and CICIDS2018 datasets as examples, we conducted visualization and analysis of the traffic data to illustrate these phenomena. The core idea of the Uniform Manifold Approximation and Projection (UMAP) technique et al. [43] is to preserve the consistency between the high-dimensional data distribution structure and the low-dimensional data distribution as much as possible. It employs a stochastic gradient descent method to identify the low-dimensional data distribution corresponding to the minimum binary cross-entropy. It can reduce the dimensionality of data for visualization purposes while maximally preserving the global structure of the data. This enables it to perform exceptionally well across various datasets. In recent years, the application of UMAP in various fields [44–46] has garnered significant attention. IoT traffic datasets are nonlinear, high-dimensional data spaces. Therefore, we utilize UMAP technology to visually analyze the distribution of some traffic in the IoTID20 and CICIDS2018 dataset, as shown in Figs. 1 and 2.

In the two-dimensional visualization of UMAP for the two datasets, the x-axis and y-axis represent two feature embeddings of traffic feature data after feature reduction by UMAP algorithm. For the IoTID20 dataset, as shown in Fig. 1(a), the blue points and pink points represent the distribution of DoS attack traffic on different dates. It can be observed that the low-dimensional visualization points of DoS attack traffic exhibit significantly different distributions. Moreover, the DoS traffic on June 4 is sparsely distributed at the boundary. This may be

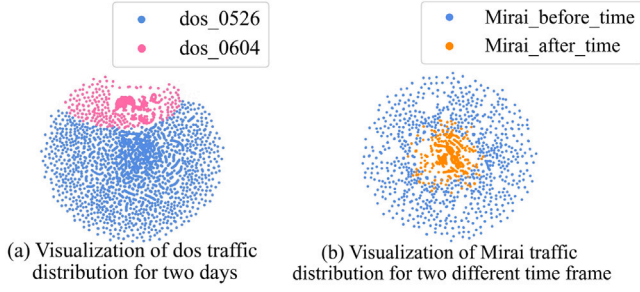


Fig. 1. Visual analysis of traffic distribution of IoTID20 based on UMAP technology.

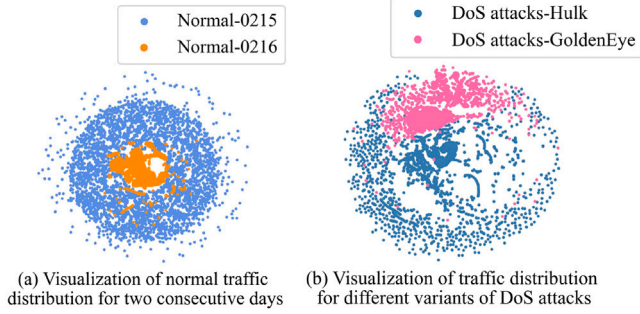


Fig. 2. Visual analysis of traffic distribution of CICIDS2018 based on UMAP technology.

due to the fact that the attacker launched DoS attacks on two different dates, resulting in changes in traffic statistical features (such as packet length, etc.). UMAP projects these traffic flows with changed characteristics to different positions, leading to a concept drift situation. At the same time, the intensity of the DoS attack traffic launched by the attacker on the two dates differed, causing the two-dimensional representation of the attack traffic on May 26 to be relatively more concentrated, while the two-dimensional representation of the DoS attack traffic on June 4 was relatively sparser. Similarly, as shown in Fig. 1(b), the blue and orange points represent the distribution of Mirai traffic with different subclasses at different times. There is a certain deviation in the distribution of Mirai traffic, and the centroids of the distributions vary. This indicates that Mirai attack traffic is also changing dynamically.

For the CICIDS2018 dataset, as shown in Fig. 2(a), the blue and orange points represent the normal traffic distributions for two consecutive days. It can be observed that there is a clear boundary between the normal traffic distributions of the two days, which is concept drift. This could be attributed to the fact that the normal traffic on the two consecutive days was influenced by different user activities or varying network behaviors, leading to discrepancies in certain traffic feature values. As a result, after dimensionality reduction using UMAP, the distribution in the two-dimensional visualization space shows notable changes. In the traffic data represented by blue dots, the traffic distribution at the edge is also discrete and sparse. Fig. 2(a) illustrates two different variants of DoS attacks — DoS attacks-Hulk and DoS attacks-GoldenEye, which also exhibit distinct boundaries and variations in the distribution of traffic feature data, that is, traffic concept drift presented by the same attack family. The distribution of DoS attack traffic exhibits both sparse and concentrated characteristics.

The above content provides a visual analysis of IoT traffic data distribution. We summarize the challenge of sparse traffic distribution: in the high-dimensional feature space, data points can be distant, causing some to be on the periphery and resulting in sparse feature representations. We also categorize the situations that trigger changes in IoT network traffic distribution into two types:

(1) Class evolution of normal traffic: Drift samples are still normal traffic samples; however, their behavioral patterns differ significantly from those of the normal traffic in the training set.

(2) Evolution of attack traffic: Attackers continuously refine and adjust their strategies, resulting in the emergence of new attack patterns derived from existing attack types, which leads to substantial changes in the characteristics of traffic data.

Concept drift: given t_0 and t_1 , $P_{t_0}(S_c)$ and $P_{t_1}(S_c)$ represent probability distribution of data samples for flow category c at times t_0 and t_1 , respectively. If $P_{t_0}(S_c) \neq P_{t_1}(S_c)$, The concept drift of IoT network traffic occurs.

The sparse distribution of traffic data makes it difficult for foundational learning models to capture effective features and patterns during the training process, leading to erroneous discrimination results. It results in a higher false positive rate and significantly reduces the detection rate. Furthermore, the drift phenomenon in traffic data means that new samples may fall outside the detection boundaries set by static learning models. When new samples arrive, the predictive capability of models built on old training data becomes severely limited, resulting in a large number of false positives. Therefore, there is an urgent need to explore more flexible and adaptive models to address the issues of data sparsity and adversarial drift, enhance the accuracy and reliability of traffic data analysis.

The research problem of our paper primarily arises from the difficulty traditional one-time learning models have in adapting to the **high-dimensional, sparse, and constantly evolving** traffic data in IoT intrusion detection. Therefore, we aim to design an adaptive IoT intrusion detection framework that can effectively accommodate the characteristics of sparse traffic distribution and concept drift in IoT networks. The framework needs to possess the following capabilities: first, it must stably and accurately learn traffic features; second, it should continuously update and adjust its detection model based on dynamically changing traffic data, thereby effectively distinguishing between normal and attack traffic. The intuitive question we need to solve is: *how can we achieve an efficient and accurate detection framework for IoT network intrusions in the context of changing traffic data distributions?*

In open environments of IoT networks, the dynamic changes in traffic distribution are an objective reality. The sparse distribution and drift of traffic samples pose significant challenges for adequate preparation and comprehensive labeling, making them unsuitable for direct training. Consequently, our research does not emphasize the use of fully prepared and fully labeled traffic data to tackle concept drift in IoT intrusion detection. Instead, it focuses on constructing a series of basic detection models that can effectively differentiate between normal and attack traffic in real-world IoT scenarios. These models enhance the generalization ability of the detection framework in IoT intrusion detection scenarios through the processes of detection, accumulation, and incremental updating of traffic data.

3.2. Framework design

In this article, we propose an incremental contrast learning-based intrusion detection framework for IoT networks. As shown in Fig. 3, our CAEAID framework can be deployed on cloud servers in the IoT. It monitors incoming traffic. These network traffic requests are captured, processed, and fed into CAEAID framework for anomaly detection. Traffic carrying malicious requests and attempts is rejected. For legitimate requests, the cloud server generates instructions or feedback that are sent to the gateway nodes and further forwarded to the corresponding IoT devices to provide normal services.

Further, we show the various steps of CAEAID framework in Fig. 4. It includes IoT raw network traffic packet capture, feature extraction and preprocessing, training, and fitting of detection models, data enhancement based on output pseudo-tags of detection model, model

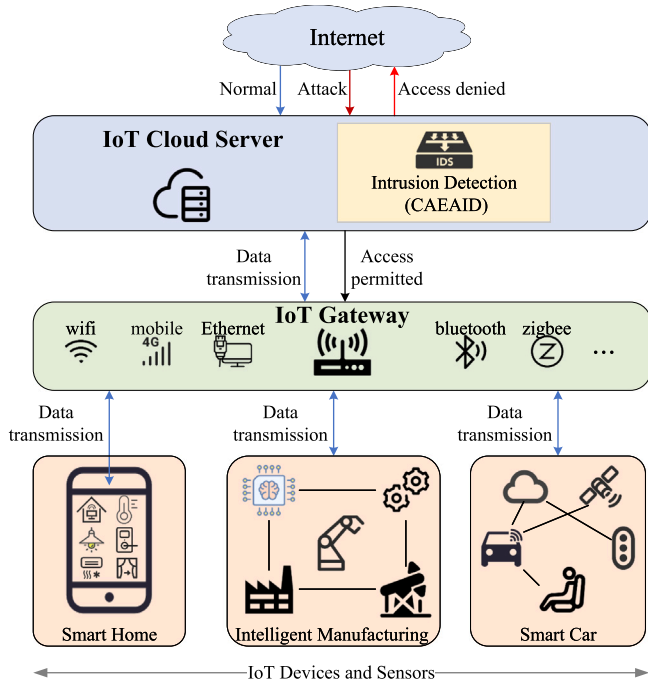


Fig. 3. Application architecture of CAEAID in IoT network intrusion detection.

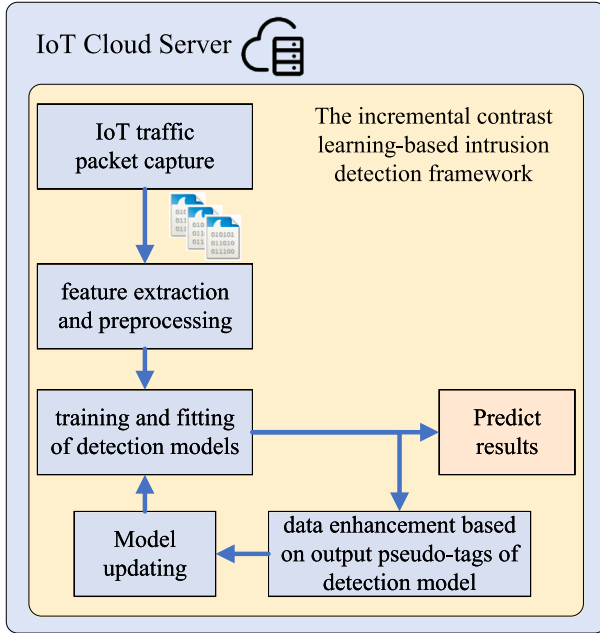


Fig. 4. The workflow of the incremental contrast learning-based intrusion detection framework for IoT network.

updating, and detection result output using learning models for detection.

The network packets captured from the IoT network are aggregated into flows. After CICFlowMeter performs statistical feature extraction based on five-tuples, we preprocess the traffic feature data. The preprocessing aims to facilitate better understanding and processing of data by the learning models and eliminate the dimensionality effects among flow features. After the above steps, the traffic characteristic dataset is obtained. The training set primarily consists of statistically aggregated features extracted from initially labeled packets. The testing set mainly

comprises statistically aggregated features extracted from subsequent unlabeled packets in the flows.

Our research mainly aims to detect intrusions suffered by IoT networks and mitigate the impact of concept drift on detection models. This section focuses on our intrusion detection framework, as shown in Fig. 5. The detection framework is mainly divided into two phases:

Phase 1 (Network Data Modeling and Detection): In this phase, we begin by analyzing and reducing the traffic feature data in the training set through the integration of kernel density estimation and box plot. Ultimately, we identify 26 key features that significantly contribute to IoT intrusion detection for the training of the learning model.

For the training of the 26-dimensional traffic feature data, we construct a contrastive autoencoder. By leveraging the available labels, we consistently minimize the distance between the low-dimensional deep feature representations of instances belonging to the same class while maximizing the distance between those of instances from different classes. Concurrently, we reduce the discrepancy between the input traffic samples and their corresponding reconstructed data produced by the autoencoder. The trained model effectively generates low-dimensional feature representations for any given input traffic sample and mitigates the impact of high-dimensional sparse data. For the testing data, we also apply feature reduction. Subsequently, we feed the feature-reduced testing set into the contrastive autoencoder model to derive low-dimensional representations. Utilizing relatively compact low-dimensional potential spaces within different classes, we develop a distance function to assign detection labels to the testing data. The contrastive autoencoder primarily emphasizes the discriminative universal features present in various classes of traffic data. However, additional attention and processing are necessary for addressing sparse extreme values. Extreme value theory is particularly adept at focusing on tail information and extreme values within the data, making it suitable for adapting to continuously changing datasets. Consequently, we extract normal traffic training samples along with the complete set of testing samples, which are then input into an enhanced extreme value theory framework for tail risk analysis. The resulting output serves as detection labels for the testing samples, thereby providing supplementary analysis for the intrusion detection model based on the contrastive autoencoder and improving the stability of the basic detection model.

Phase 2 (Data Augmentation and Model Incremental Learning): The CAEAID framework is designed to effectively detect intrusion behaviors, such as attacks, within the expanding IoT network. Consequently, the training process of the framework must learn from new data while preserving historical knowledge. In this phase, it is essential to combine the testing results of the detection model to facilitate data augmentation and model incremental learning. We assess the consistency of the detection labels provided by the contrastive autoencoder and the improved extreme value theory. We identify testing samples with consistent detection labels and utilize this subset of samples for retraining and updating the model in Phase 1, enabling the model to possess incremental learning capabilities. It improves adaptability to concept drift in intrusion detection and enhances the model's performance in IoT network intrusion detection.

Additionally, we explore the potential applications of the CAEAID framework in real-world IoT scenarios. Taking smart homes as an example, with the increasing prevalence of smart home devices, home networks face significant security challenges. By deploying the CAEAID framework on cloud servers, we can monitor the network traffic between smart home devices in real-time, promptly identifying abnormal behaviors and potential threats. Furthermore, the incremental learning capability of the CAEAID framework allows it to gradually adapt to new patterns that may emerge in smart home traffic, maintaining efficient detection performance. The CAEAID framework can also collaborate with other security strategies, such as access control mechanisms. By leveraging the abnormal traffic detected by the CAEAID framework, access control mechanisms can block the IP addresses associated with such traffic to prevent potential attacks or data breaches, thereby enhancing the overall security of smart home environments.

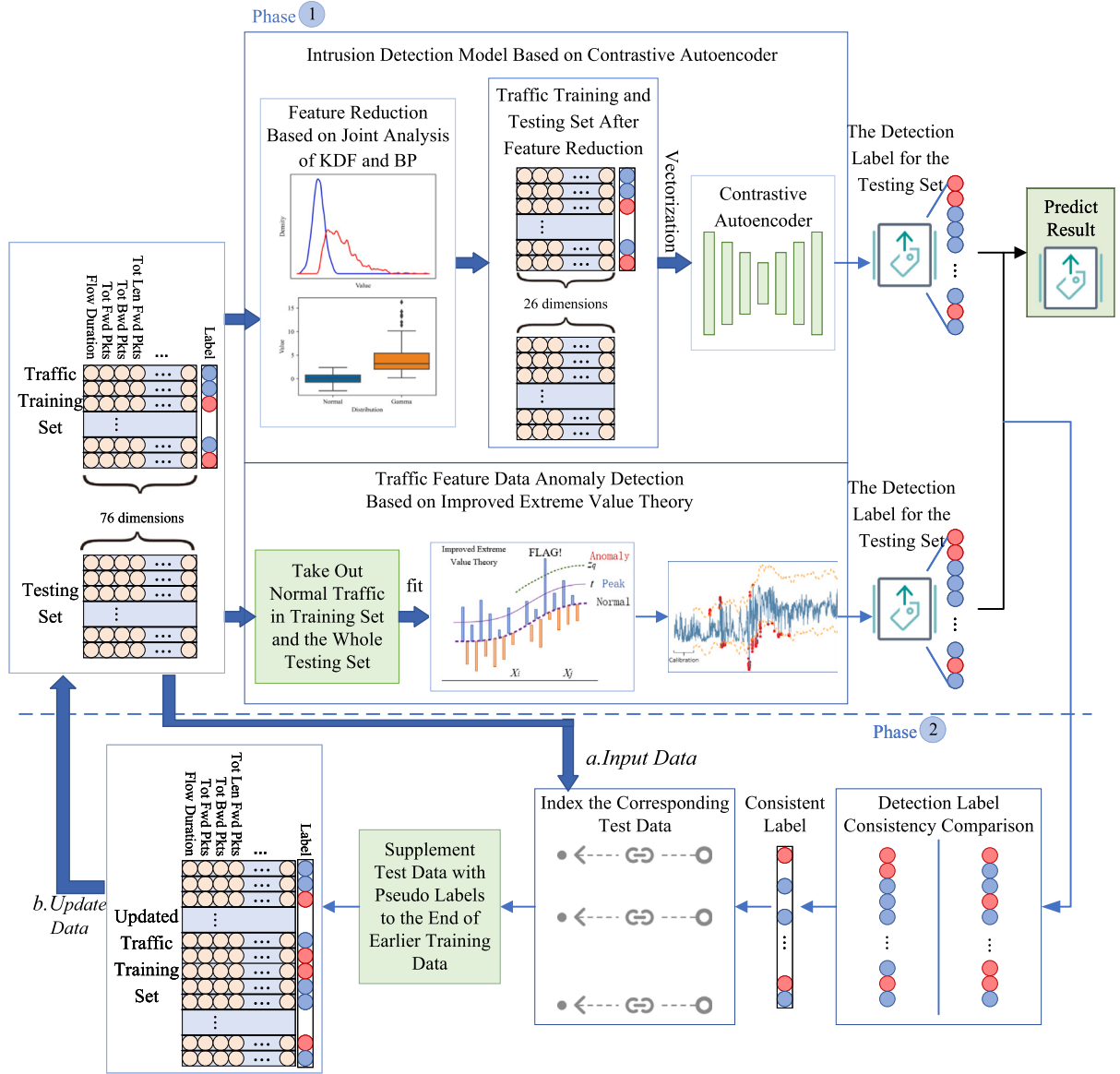


Fig. 5. CAEAID Framework.

3.3. Implementation details

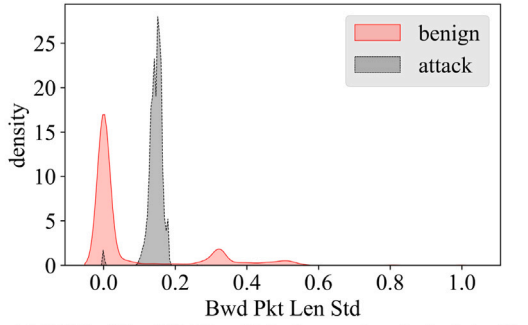
This section mainly discusses the implementation details of two stages in the incremental contrast learning-based intrusion detection framework for IoT networks.

3.3.1. Intrusion detection model based on contrastive autoencoder

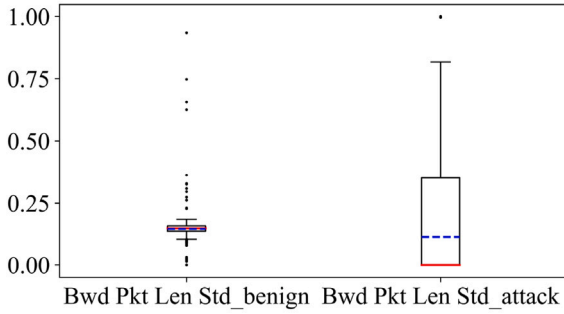
After traffic statistics feature extraction, label encoding, data normalization, and other pre-processing operations, each traffic data instance contains 76-dimensional features. Due to the high dimension of traffic statistics features, there are some redundant features that contribute little to the IoT intrusion detection model based on DL learning, and the distribution difference between different traffic types is fuzzy. These features may interfere with the learning process of the detection model or cause the model to bias some specific outputs. It causes the model to overlook important feature patterns during the training process, negatively impacting detection performance. Therefore, it is necessary to remove the redundancy, low frequency, or features that contribute little to IoT network intrusion detection, and improve the generalization and training efficiency of the detection model.

For feature reduction, we need to analyze the correlation between features and traffic labels, assessing the impact of features on distinguishing IoT traffic to select key features. The kernel density plot and box plot offer distinct perspectives on the distribution of data features. The kernel density plot presents a smooth probability density function, facilitating an understanding of the distribution's shape, while the box plot conveys crucial information, such as the median and quartiles, which are essential for assessing central tendency and dispersion. By integrating these two types of visualizations, we can more comprehensively evaluate the potential impact of each feature on the attack category.

For instance, Fig. 6(a) displays the kernel density plot for the Backward Packet Length Standard Deviation, referred to as Bwd Pkt Len Std, for both normal and attack traffic. The red solid line represents the Bwd Pkt Len Std for normal traffic, whereas the black dashed line corresponds to the Bwd Pkt Len Std for attack traffic. The Backward Packet Length Standard Deviation shows distinctly different distributions for normal and attack traffic in the kernel density plot. Fig. 6(b) presents the box plot of Bwd Pkt Len Std for normal and attack traffic, with the left box plot depicting normal traffic and the right box plot illustrating attack traffic. It is apparent that this feature



(a) KDF of Bwd Pkt Len Std of normal and attack traffic



(b) BP of Bwd Pkt Len Std of normal and attack traffic

Fig. 6. Comparison of kernel density plot and box plot on the Bwd Pkt Len Std of normal traffic and attack traffic.

demonstrates varying medians and interquartile ranges for normal and attack traffic. Collectively, these two plots suggest that this feature may possess significant discriminative power for the traffic detection task.

Due to space constraints, we do not present the comparison of kernel density plots and box plots for all flow characteristics in this paper. We conducted a joint analysis of 76-dimensional traffic features by kernel density plot and box plot. Finally, we determined that the 26-dimensional traffic features with high differentiation between normal and attack traffic are as follows: 'Flow Duration', 'Fwd Pkt Len Max', 'Fwd Pkt Len Mean', 'Fwd Pkt Len Std', 'Flow IAT Max', 'Fwd IAT Std', 'Pkt Len Max', 'Fwd Seg Size Avg', 'Bwd Pkt Len Max', 'Bwd Pkt Len Mean', 'Bwd Pkt Len Std', 'Flow IAT Std', 'Flow IAT Max', 'Fwd IAT Tot', 'Fwd IAT Std', 'Fwd IAT Max', 'Pkt Len Max', 'Pkt Len Mean', 'Pkt Len Std', 'Pkt Len Var', 'RST Flag Cnt', 'ECE Flag Cnt', 'Pkt Size Avg', 'Fwd Seg Size Avg', 'Bwd Seg Size Avg', 'Init Fwd Win Byts'. The remaining features exhibit a certain degree of overlap in their distribution between normal traffic and attack traffic, which may be detrimental to the subsequent model learning process. We use the traffic data with these 26-dimensional characteristics to train the learning model.

According to the literature [47], calculating the distance function between traffic samples and existing categories can effectively mitigate concept drift behavior. Consequently, we investigate the concept of distance comparison for intrusion detection. Traffic samples that demonstrate concept drift behavior may reside in sparse outlier spaces, where high-dimensional sparse data can hinder the effectiveness of the distance function. Therefore, to construct an effective distance function, learning a low-dimensional feature representation of traffic samples and clearly defining the boundaries between categories is essential.

In this regard, we introduce a contrastive autoencoder designed to learn meaningful and robust feature representations of traffic data. Contrastive learning emphasizes the relative differences between data instances, thereby improving the generalization capabilities of classification and detection. It is achieved by continuously promoting closer representations of similar samples in the feature space while pushing

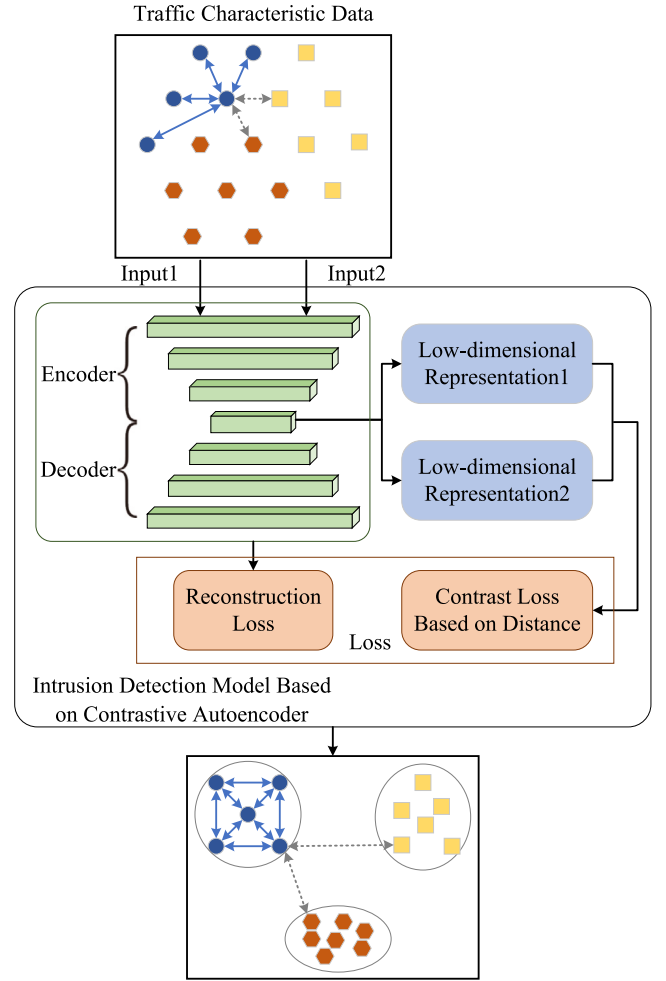


Fig. 7. Contrastive Autoencoder Model.

the representations of dissimilar samples further apart. Autoencoders, which learn a compressed representation of data and reconstruct the input, offer significant advantages in acquiring low-dimensional latent feature representations. By integrating the principles of contrastive learning with autoencoders, we aim to extract key features from traffic data and enhance the model's sensitivity to the distinctions between normal and attack traffic.

We propose an intrusion detection model based on a contrastive autoencoder, as illustrated in Fig. 7. Given high-dimensional input traffic feature instances, the autoencoder maps these instances into a low-dimensional latent feature space. We incorporate contrastive learning to minimize the distance between similar samples while maximizing the distance between different types of samples, thereby enhancing the distinctiveness of the low-dimensional feature representations for each traffic category. Specifically, the contrastive autoencoder processes pairs of traffic instances, which may consist of pairs of normal traffic samples, pairs of normal and attack traffic samples, or pairs of attack traffic samples. During the encoding phase, the encoder transforms the traffic samples into low-dimensional latent space representations. It aims to capture the essential features of the traffic samples. The encoding process for obtaining the low-dimensional latent feature representations is shown in Eq. (1):

$$z = g_{\theta_1}(x) \quad (1)$$

During the decoding phase, the decoder attempts to reconstruct the original input traffic from the latent space representation of the traffic data, helping to ensure that the learned feature representations can

retain important information from the input traffic feature data. The decoding process of the reconstruction phase is shown in Eq. (2):

$$\hat{x} = h_{\theta_2}(z) = h_{\theta_2}(g_{\theta_1}(x)) \quad (2)$$

The contrastive autoencoder's loss consists of reconstruction loss and contrastive loss, as shown in Eq. (3):

$$Loss = Reconstruct_Loss + \lambda \cdot Contrast_Loss$$

$$= \frac{1}{N} \sum_{k=1}^N \|x - \hat{x}\| + \lambda \left(\frac{1}{2N} \sum_{k=1}^N y_{ij} d_{ij}^2 + \max(\text{margin} - d_{ij}, 0)^2 \right) \quad (3)$$

where the first term is the reconstruction loss of the autoencoder, which represents the mean squared error between the input traffic sample x and its reconstruction $\hat{x}$ by the autoencoder; N represents the number of samples. The second term is the contrastive loss, which measures the degree of match between paired samples. In the contrastive loss, the margin is a threshold, which is set to 1. The variable λ controls the weight of the contrastive loss, where $0 < \lambda < 1$, and y_{ij} indicates whether a pair of samples belong to the same class. The calculation formula for y_{ij} is shown in Eq. (4):

$$y_{ij} = \begin{cases} 0, & \text{if } x_i \in C_k \text{ and } x_j \notin C_k \\ 1, & \text{if } x_i \in C_k \text{ and } x_j \in C_k \end{cases} \quad (4)$$

Also, d_{ij} indicates the Euclidean distance, 2-Norm, of the low-dimensional representation z_i, z_j of a pair of input traffic samples. Its formula is shown in Eq. (5):

$$d_{ij} = \|z_i - z_j\|_2 = \left(\sum_{m=1}^M (z_i^m - z_j^m)^2 \right)^{\frac{1}{2}} \quad (5)$$

where M represents the number of dimensions of low-dimensional embeddings. The contrastive autoencoder is trained by continuously minimizing the loss function, which effectively maps traffic instances to a low-dimensional latent feature space. Similar traffic data are closer together, while different types of traffic data are more distant. After training, we further design a distance function to detect subsequent traffic data.

For a traffic sample to be tested x_j^{test} , we determine its category by construct a distance function between x_j^{test} and the known, accurately labeled traffic data. Suppose there are T categories in the training samples, with each category containing s_i samples, where $i = 1, 2, \dots, T$. For the i th category of training samples, we input them into the contrastive autoencoder for mapping, aggregating them into a low-dimensional latent feature space $(z_1, z_2, \dots, z_{s_i})$. Then, we calculate the centroid of each category C_i , with the calculation formula as shown in Eq. (6):

$$c_i = \frac{(z_1 + z_2 + \dots + z_{s_i})}{s_i} \quad (6)$$

Then, for each category of traffic training samples, we calculate the Mean Intra-cluster Distance (MID) of their low-dimensional representations. Its calculation formula is shown in Eq. (7):

$$MID_i = \text{mean}(\|z_1 - c_i\|, \dots, \|z_{s_i} - c_i\|) \quad (7)$$

For x_j^{test} , we input it into the contrastive autoencoder to obtain its low-dimensional latent feature representation x_j^{test} and calculate its distance to the centroids c_i of the normal and attack traffic training samples. Some testing traffic samples may have distances to the centroid that are inadequate for accurately indicating their proximity to the traffic category boundary. To improve the detection method's capacity to differentiate between normal and attack traffic latent representations, we classify the testing sample based on the difference between its distance to each category's low-dimensional representation centroid and the average intra-cluster distance of that category's low-dimensional representation. The category that is closest to the boundary is then selected as the classification for the testing traffic. The formula for determining the category of the testing traffic sample

Algorithm 1 The detection process of the contrastive autoencoder model.

Input: Traffic training data $X_{train} = \{x_{ij} | i = 1, \dots, T, j = 1, \dots, s_i\}$;
The number of traffic classes T ;
Sample size of the i -th category in traffic s_i ;
Traffic testing data $X_{test} = \{x_{t_m} | m = 1, \dots, M\}$;
The number of testing traffic samples M .

Output: Predict label $Y_{test} = \{y_{t_m} | m = 1, \dots, M\}$.

```

1: for  $i = 1$  to  $T$  do
2:   for  $j = 1$  to  $s_i$  do
3:      $z_{ij} = g_{\theta_1}(x_{ij})$ 
4:   end for
5:   The centroid of each category  $c_i = \frac{(z_{i1} + z_{i2} + \dots + z_{is_i})}{s_i}$ 
6:   for  $j = 1$  to  $s_i$  do
7:     Calculate the distance from each sample in the  $i$ th category
       to its centroid  $d_{ij} = \|z_{ij} - c_i\|$ 
8:   end for
9:   Mean Intra-cluster Distance of each category  $MID_i =$ 
      $\text{mean}(d_{i1}, d_{i2}, \dots, d_{is_i})$ 
10: end for
11: for  $m = 1$  to  $M$  do
12:    $z_{t_m} = g_{\theta_1}(x_{t_m})$ 
13:   for  $i = 1$  to  $T$  do
14:      $d_{im} = \|z_{t_m} - c_i\|$ 
15:   end for
16:    $y_{t_m} = \arg \min_n (\|z - c_i\| - MID_i)$ 
17: end for

```

is shown in Eq. (8):

$$y_j = \arg \min_n (\|z_j^{test} - c_i\| - MID_i) \quad (8)$$

The detection process of the contrastive autoencoder model is shown in Algorithm 1. Lines 1–10 represent the procedure for calculating the average cohesion distance of the low-dimensional representation of traffic for each category. Lines 10–17 indicate the classification of the traffic to be measured according to the distance between the low-dimensional representation of the data to be measured and the low-dimensional representation of the centroid of each type of traffic.

3.3.2. Anomaly detection of traffic characteristics based on improved extreme value theory

With the arrival of new traffic data, it is imperative to update the model. We must effectively leverage new traffic data to enhance its generalization capability. To increase the accuracy of the pseudo-labels for the traffic data utilized in updating the contrastive autoencoders, we improve extreme value theory to analyze the comprehensive statistical characteristics of traffic and differentiate between normal and attack traffic. Additionally, we will perform a consistency analysis between the results of this analysis and the detection outcomes of the intrusion detection model based on contrastive autoencoder to establish the pseudo-labels for the traffic data used in model updates.

According to the box plot analysis of traffic statistical features in Section 3.3.1, it can be observed that many traffic feature data contain some outliers or extreme values. Since contrastive autoencoders primarily focus on learning the general feature representations of the data, they may not directly pay attention to the tail information, making it difficult to identify and handle extreme values effectively. Extreme value theory is particularly effective in fields that focus on the tail behavior of data. Therefore, we introduce extreme value theory into analyzing the full statistical features of traffic to identify normal and attack traffic.

Extreme value theory can be applied without any prior assumptions regarding the traffic data and has shown excellent scalability

in managing large-scale, high-dimensional datasets. Specifically, for a given sequence of traffic feature data, we first group the normal traffic training data to form the initial segment. Using the traffic feature data from these initial segments, we establish the initial peak threshold and calculate the initial anomaly threshold through extreme value theory. During the subsequent sliding window process, we assess whether the values of the traffic feature data in every sliding window surpass the anomaly threshold. If it is found that a traffic characteristic sample exceeds the anomaly threshold, the sample will be marked as an anomaly sample. It is removed from the sliding window to prevent any distortion of the traffic feature data sequence distribution within the window. If the traffic feature value only exceeds the initial peak threshold, it indicates that concept drift has occurred in the sequence. In this case, the extreme value distribution needs to be updated, the traffic feature data needs to be refitted, and the anomaly threshold should be updated based on the new data. If the traffic feature data is determined to be normal, the threshold remains unchanged, and the system continues to monitor new traffic inputs.

Stream Peak Over Thresholding (SPOT) is an anomaly detection algorithm for streaming data based on extreme value theory, explicitly utilizing the second theorem of extreme value theory. The determination of the anomaly threshold in SPOT relies on the distribution of extremes, and the extreme value of time series traffic feature data satisfies the generalized Pareto distribution, with its cumulative function given by Eq. (9):

$$\frac{\bar{F}(t + \sigma(t)x)}{\bar{F}(t)} \xrightarrow{t \rightarrow \tau} (1 + \gamma x)^{-\frac{1}{\gamma}} \quad (9)$$

where t is the initial threshold for the anomaly fraction that detects the peak, γ and σ are the parameters of the generalized Pareto distribution that needs to be evaluated. γ and σ are solved based on optimization algorithms such as the Probability Weighted Moments or the Method of Moments. After that, the target threshold can be obtained z_q . When $X > z_q$, the value is considered as an abnormal traffic data point.

However, the SPOT method operates under the assumption that the distribution of X remains constant over time. It renders SPOT ineffective in detecting local peaks when encountering concept drift within traffic data. To address this limitation, we propose an adaptive stream peak-over-threshold algorithm for detecting anomalies in IoT network traffic. Unlike the SPOT algorithm, we set $|X - t|$ as the portion exceeding the initial threshold t , which allows us to establish both upper and lower thresholds. Also, we overcome the limitations of SPOT by modeling the average local behavior and applying the SPOT algorithm on relative gaps. The improved cumulative formula is shown in Eq. (10):

$$\bar{F}_t(x) = P(|X - t| > x | X > t) \xrightarrow{t \rightarrow \tau} \left(1 + \frac{\gamma x}{\sigma(t)}\right)^{-\frac{1}{\gamma}} \quad (10)$$

To facilitate the inference process, the upper and lower thresholds for t are set at the 98th and 2nd percentiles, respectively. The thresholds are dynamically determined in (11)–(12):

$$t_A^\mu \simeq t^\mu + \frac{\hat{\sigma}}{\hat{\gamma}} \left(\left(\frac{qN}{N_t} \right)^{-\hat{\gamma}} - 1 \right) \quad (11)$$

$$t_A^l \simeq t^l - \frac{\hat{\sigma}}{\hat{\gamma}} \left(\left(\frac{qN}{N_t} \right)^{-\hat{\gamma}} - 1 \right) \quad (12)$$

where $\hat{\sigma}$ and $\hat{\gamma}$ are the evaluation parameter, N_t is the number of peaks, N is the number of abnormal scores in the sliding window, q is the risk level which determine anomaly. $\hat{\sigma}$ and $\hat{\gamma}$ can be calculated by maximum likelihood estimation. Their calculation formula is shown in Eqs. (13)–(14):

$$\hat{\sigma} = \frac{\mu}{2} \left(1 + \frac{\mu^2}{S^2} \right) \quad (13)$$

Algorithm 2 Anomaly detection algorithm of flow characteristics based on improved extreme value theory.

Input: Traffic characteristic sequence $T = [S_1^T, \dots, S_m^T]$;
the number of traffic features m , $S_i = [S_{i1}, \dots, S_{in}]$;
the length of sliding window ω ;
the length of the check sequence l ;
probability of abnormal risk q .

Output: Anomaly labels $Y_{EVT} = [y_1, y_2, \dots, y_n]$.

```

1: for  $i$  in  $m$  do
2:    $E_i^* = [S_{i1}, \dots, S_{id}]$ 
3:    $M_{i,d+1} \leftarrow \bar{E}_i^*$ 
4:   for  $j$  in  $[d+1, d+l]$  do
5:      $S'_{ij} \leftarrow S_{ij} - M_{ij}$ 
6:      $E_i^* \leftarrow E_i^*[1:] \cup S_{ij}$ 
7:      $M_{i,j+1} \leftarrow \bar{E}_i^*$ 
8:   end for
9:    $t_i^{upper}, t_i^{lower} \leftarrow \text{SetPeakT}(E_i^*)$ 
10:   $L_{it} \leftarrow \{S_{ij} - t_i^{upper} | S_{ij} > t_i^{upper}\} \cup \{t_i^{lower} - S_{ij} | S_{ij} < t_i^{lower}\}, S_{ij} \in E_i^*$ 
11:   $N_{it} = |L_{it}|$ 
12:   $\hat{\gamma}_i, \hat{\sigma}_i \leftarrow \text{MOM}(L_{it})$ 
13:   $t_{iA} \leftarrow \text{CalculateT}(q, \hat{\gamma}_i, \hat{\sigma}_i, L_{it}, l, N_{it}, \omega)$ 
14:  for  $j$  in  $[d+l+1, n]$  do
15:     $y_{ij} \leftarrow 0$ 
16:     $S'_{ij} \leftarrow S_{ij} - M_{ij}$ 
17:    if  $S'_{ij} > t_{iA}^{upper}$  or  $S'_{ij} < t_{iA}^{lower}$  then
18:       $y_{ij} \leftarrow 1$ 
19:       $M_{i,j+1} \leftarrow M_{ij}$ 
20:    else if  $S'_{ij} > t_i^{upper}$  or  $S'_{ij} < t_i^{lower}$  then
21:       $L_{ij} \leftarrow S'_{ij} - t_i^{upper}$  or  $L_{ij} \leftarrow t_i^{lower} - S'_{ij}$ 
22:      Add  $L_{ij}$  to  $L_{it}$ 
23:       $l \leftarrow l + 1$ 
24:       $N_{it} \leftarrow N_{it} + 1$ 
25:       $\hat{\gamma}_i, \hat{\sigma}_i \leftarrow \text{MOM}(L_{it})$ 
26:       $t_{iA} \leftarrow \text{CalculateT}(q, \hat{\gamma}_i, \hat{\sigma}_i, L_{it}, l, N_{it}, \omega)$ 
27:       $E_i^* \leftarrow E_i^*[1:] \cup S_{ij}$ 
28:       $t_i \leftarrow \text{SetPeakT}(E_i^*)$ 
29:       $M_{i,j+1} \leftarrow \bar{E}_i^*$ 
30:    else
31:       $l \leftarrow l + 1$ 
32:       $E_i^* \leftarrow E_i^*[1:] \cup S_{ij}$ 
33:       $M_{i,j+1} \leftarrow \bar{E}_i^*$ 
34:    end if
35:  end for
36:   $y_j = y_{ij} | y_{i-1,j}$ 
37: end for
38: Return  $Y_{EVT}$ 

```

$$\hat{\gamma} = \frac{1}{2} \left(1 - \frac{\mu^2}{S^2} \right) \quad (14)$$

$$\text{where } \mu = \frac{\sigma}{1-\gamma}, S^2 = \frac{\sigma^2}{(1-\gamma)^2(1-2\gamma)}.$$

Anomaly detection algorithm of flow characteristics based on improved extreme value theory is shown in Algorithm 2.

3.3.3. Concept drift adaptation and model incremental learning

The previous subsections primarily focus on obtaining a comprehensive, accurate low-dimensional representation of the model from the perspective of basic model design. They ensure that the model establishes clear category boundaries and generates relatively more precise pseudo-labels and alleviates the impact of high-dimensional discrete data on the detection model. To further adapt to the dynamic changes inherent in traffic data within IoT networks, and to enhance the model's generalization capabilities for detection, it is essential to fully leverage

Table 1

Hyperparameter settings of contrastive autoencoder.

Hyperparameter settings	Value
Batch size	256
Learning rate	0.001
Optimizer	Adam
Epoch	250

new unlabeled testing data for dynamic model adjustments, thereby endowing it with incremental learning capabilities. To this end, we periodically retrain the model using both new and old data in the context of concept drift adaptation, effectively replacing the outdated model. To ascertain pseudo-labels for the new data, we must jointly evaluate the normal and attack detection outcomes provided by the intrusion detection model, which is based on a contrastive autoencoder, alongside the anomaly detection of traffic characteristics utilizing an improved extreme value theory. We extract results that are consistent across both detection algorithms and index the corresponding testing traffic data according to their sequence numbers. These detection results are then employed as pseudo-labels, augmenting the existing traffic training data to facilitate data enhancement. Subsequently, we input the updated training data back into the detection model for retraining, leading to model updates that improve detection performance and adapt to concept drift. Finally, we present the detection results, while the outcomes from the contrastive autoencoder remain unchanged.

For example, given seven testing samples $x_1, x_2, x_3, x_4, x_5, x_6, x_7$, the detection results obtained from the intrusion detection model based on contrastive autoencoders are [0, 0, 1, 1, 1, 0, 1]. Simultaneously, when inputting the same samples into the anomaly detection of traffic characteristics based on improved extreme value theory, the detection results are [0, 1, 1, 1, 0, 0, 1]. By comparing the consistency of the detection results, we find that the traffic samples with consistent detection results are x_1, x_3, x_4, x_6, x_7 , and the pseudo-labels corresponding to these five samples are [0, 1, 1, 0, 1]. We then supplement these five testing samples to the original training data X_{train} for data augmentation and model updates. When the model's detection results remain stable, the final detection results for the testing samples are produced.

4. Experimental evaluation and analysis

4.1. Experimental setup and evaluation metrics

Experimental Setting. Our experimental environment is set up as follows: Windows 10 operating system, Intel(R) Core (TM) i7-8700 CPU with 32.0 GB RAM, and NVIDIA GeForce RTX 2060 GPU. We implemented the algorithm using Python 3.7 and TensorFlow.

In Section 3.3.1, the internal structure of the encoder and decoder of our contrastive autoencoder is set as fully connected layers. The number of neurons in each layer of the encoder is 26-16-8-3, while the number of neurons in each layer of the decoder is 3-8-16-26. The performance of the contrastive autoencoder is influenced by hyperparameters, and our hyperparameter settings for the contrastive autoencoder are shown in Table 1. The loss variations of the model on the IoTID20 and CICIDS2018 datasets are illustrated in Fig. 8. It is evident that the epochs stabilize after 250 for both datasets; therefore, we set the epoch of the contrastive autoencoder to 250.

Datasets. After extensive literature review [6,34,48], our study utilized two popular available datasets, IoTID20 and CICIDS2018, to evaluate our framework.

- The IoTID20 and CICIDS2018 datasets encompass a variety of attack types, offering diversity and richness in attacks, which enables us to more comprehensively evaluate the performance of the CAEAD framework in IoT intrusion detection.

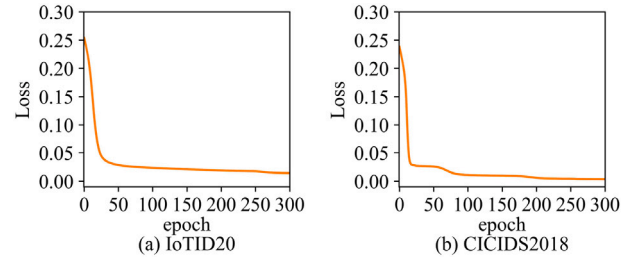


Fig. 8. Loss variations of the contrastive autoencoder on the IoTID20 and CICIDS2018 training datasets.

Table 2

Traffic data distribution of IoTID20.

	Category	Family	Number of samples
Initial training set	Normal	Benign	32058
	Attack	DoS	44666
		Mirai	29450
Initial testing set	Normal	Benign	8015
	Attack	DoS	14725
		Mirai	12118

- These two datasets are relatively recent. Given the rapid development of IoT technology, new attack methods and strategies are constantly emerging. These datasets can more effectively simulate the security threats in IoT networks.
- The IoTID20 and CICIDS2018 datasets have been widely used in multiple papers on IoT intrusion detection [6,34,48,49]. This trend indicates the recognition and reliability of these datasets in the academic community. By using these datasets, we can more conveniently compare our work with others and evaluate the performance of different intrusion detection algorithms.
- Through the analysis in Section 3, these two datasets also present the issues of concept drift and sparse data distribution quite well, which can effectively assist us in assessing and validating the adaptability of our framework in the face of these two problems.

The IoTID20 dataset consists of normal and network traffic data generated by attacking IoT devices. This dataset was constructed by Ullah et al. [23] using Cicflowmeter to extract features such as flow duration, total forwarded packets, activity, and idle time from the network traffic.

The CICIDS2018 dataset is a well-annotated and relatively new dataset with a wide variety of attack categories in intrusion detection. The dataset has been widely used in recent cybersecurity research for experimental evaluation and analysis of intrusion detection in various real-world scenarios. At the same time, this dataset features the latest and most sparse characteristics, making it closer to the traffic observed in real IoT environments.

To validate the generalization ability of various methods in addressing concept drift behavior within IoT network intrusion detection, we extracted a subset of the traffic data from the IoTID20 dataset for experimentation. For the construction of concept drift traffic data, we make the following selections. The attack traffic categories present in the original training data include DoS from May 26, Mirai-Ackflooding, Mirai-HTTP Flooding, and Mirai-UDP Flooding. In contrast, the testing data encompasses attack traffic categories such as DoS from May 26 and Mirai-Hostbruteforce. The normal traffic in both the training and testing sets is distributed in a ratio of 4:1. The attack traffic subclasses in the training set and the testing set are different, and the testing traffic samples do not appear in the original training traffic samples, resulting in a significant shift. The final data distribution is illustrated in Table 2.

The CICIDS2018 dataset is extensive and contains redundant samples. For our experimentation, we extracted data from this dataset to

Table 3
Traffic data distribution of CICIDS2018.

	Category	Family	Number of samples
Initial training set	Normal	Benign	257011
		DDoS	68601
	Attack	DoS	60180
		Infiltration	9240
Initial testing set	Normal	Benign	128505
		DDoS	31730
	Attack	DoS	52498
		Infiltration	38236

assess the generalization capabilities of various methods concerning concept drift in intrusion detection. For the construction of concept drift traffic data during experimental evaluation, we make the following divisions. We selected data from five specific days: February 15, February 16, February 21, February 28, and March 1, allocating 10% of the data for training purposes. The attack subclasses included in the training dataset are DDOS attack-HOIC, DoS attack-Hulk, and Infiltration. Additionally, we extracted 5% of the data for testing and classified the attack subclasses not present in the training data as drift data. The attack subclasses included in the testing traffic data are DDOS attack-LOIC-UDP, DoS attacks-GoldenEye, DoS attacks-Slowloris, and Infiltration. The final data distribution is presented in Table 3.

Baseline Studies. In this work, we compare the performance of our CAEAID framework with classical and state-of-the-art research. The compared IoT intrusion detection approaches are categorized into three groups: (1) classical ML and DL methods; (2) superior approaches in active drift detection and passive mitigation experiments for IoT intrusion detection; and (3) advanced methods in IoT intrusion detection. For classical ML/DL methods, we selected XGBoost, MiniBatchKMeans, DNN, and AE algorithms. XGBoost is relatively lightweight and exhibits better performance in machine learning. MiniBatchKMeans is also a distance-based unsupervised clustering algorithm. DNN and AE are better suited for learning the statistical features of IoT traffic data. For advanced active drift detection and passive adaptation schemes in IoT scenarios, we chose PWPAE [34], CADE [37], and ENIDrift [39]. PWPAE and CADE focus on drift detection techniques, while ENIDrift emphasizes dynamic updates to ensemble models to adapt to the evolving nature of IoT traffic data. For cutting-edge IoT intrusion detection methods, we selected FL-IIDS [28] and DWGF-IDS [27] as baseline models. We fully reproduce these learning models in our work and compare the performance with our CAEAID framework.

Evaluation Metrics. The CAEAID framework and the compared methods in experiments are all ML or DL-based IoT intrusion detection approaches. To ensure a comprehensive evaluation, we adopt multiple metrics, including confusion matrix, accuracy, precision, recall, F1-score, and false positive rate, to quantify their performance. This multi-dimensional analysis enables a thorough experimental assessment. Accuracy measures the model's overall performance in detecting various traffic categories. Precision measures the proportion of correctly identified instances among those detected as attacks. Recall assesses the proportion of correctly detected instances among those that are actually attack traffic. F1-score is the harmonic mean of precision and recall. The false positive rate measures the proportion of instances incorrectly detected among those actually normal traffic.

4.2. Experimental results and analysis

4.2.1. Comparison of state-of-the-art methods in IoT network intrusion detection

We first conduct experiments to compare the performance of our CAEAID framework with the baseline studies described in Section 4.1,

Table 4
Comparison of detection accuracy(%), precision(%), recall(%) and F1-score(%) of different methods on IoTID20.

Method	Accuracy	Precision	Recall	F1-score
XGBoost	70.51	90.84	68.62	78.18
MiniBatchKMeans	75.44	88.79	77.95	83.02
DNN	76.60	88.23	79.51	82.71
AE	57.38	76.60	65.49	70.23
PWPAE	84.92	92.48	87.53	89.93
CADE	86.04	93.24	88.27	90.69
ENIDrift	85.20	92.30	88.14	90.17
DWGF-IDS	81.89	90.94	84.95	87.85
FL-IIDS	82.81	91.83	85.26	88.42
CAEAID	87.03	93.77	89.08	91.36

including XGBoost, MiniBatchKMeans, DNN, AE, PWPAE, CADE, ENIDrift, FL-IIDS, and DWGF-IDS.

Table 4 and Fig. 9 compare the ability of different methods to distinguish between normal and attack traffic on the IoTID20 dataset. According to Table 4, the CAEAID for IoTID20 achieves an accuracy of 87.03%, a precision of 93.77%, a recall of 89.08%, and an F1-score of 91.36% when detecting traffic that includes sparse and drift samples. CAEAID achieves the best detection performance across all methods. This is attributed to its comprehensive and stable feature learning with incremental update capabilities. XGBoost, MiniBatchKMeans, DNN, and AE show the lowest accuracy among all compared methods. CAEAID improves accuracy by 13.6% compared to DNN, the top performer in this group. For attack traffic detection, CAEAID achieves at least a 12.04% higher detection rate than these four methods. This indicates that classical ML/DL methods perform poorly when handling concept drift in IoT traffic.

FL-IIDS and DWGF-IDS, as advanced IoT intrusion detection methods, outperform classical ML/DL algorithms in detection performance. However, CAEAID further improves accuracy by 5.10% and F1-score by 3.33% compared to FL-IIDS. In the field of IoT intrusion detection, advanced active drift detection, passive mitigation adaptation methods PWPAE, CADE, and ENIDrift outperform other two categories in detection performance. CAEAID improves upon the best method among others, CADE, by 1.15%. In terms of recall, CAEAID shows an improvement of 0.92% over CADE. CAEAID also achieved the highest F1-score.

According to Fig. 9, the false positive rates are high due to the low total number of normal samples in the testing set. Some normal samples can be misclassified as attack samples, contributing to high false positive rates. Therefore, the false positive rates of all methods on IoTID20 are generally high. Our framework achieved the lowest false positive rate. The above results demonstrate that our CAEAID framework can effectively mitigate and address the impact of concept drift issues in IoT intrusion detection.

We compared the detection performance of the aforementioned methods with the CAEAID framework on CICIDS2018, as illustrated in Table 5 and Fig. 10. XGBoost, MiniBatchKMeans, DNN, and AE exhibited limited generalization capabilities when confronted with new normal and attack traffic. While the XGBoost model effectively detected normal traffic, it demonstrated a low recall for new attack traffic, suggesting difficulties in adapting to emerging attack patterns. Similarly, DNN and AE, both deep learning models, tended to capture universal features of network traffic but also exhibited subpar detection performance. It indicates that classical ML and DL models struggle to accurately identify sparse and dynamic traffic.

DWGF-IDS and FL-IIDS showed improvements across various performance metrics compared to classical ML and DL methods. Specifically, FL-IIDS enhanced accuracy and F1-score by 15.25% and 33.6%, respectively, compared to XGBoost. Additionally, PWPAE, CADE, and ENIDrift demonstrated significant performance gains over FL-IIDS and DWGF-IDS, achieving a minimum increase of 2.89% in accuracy and

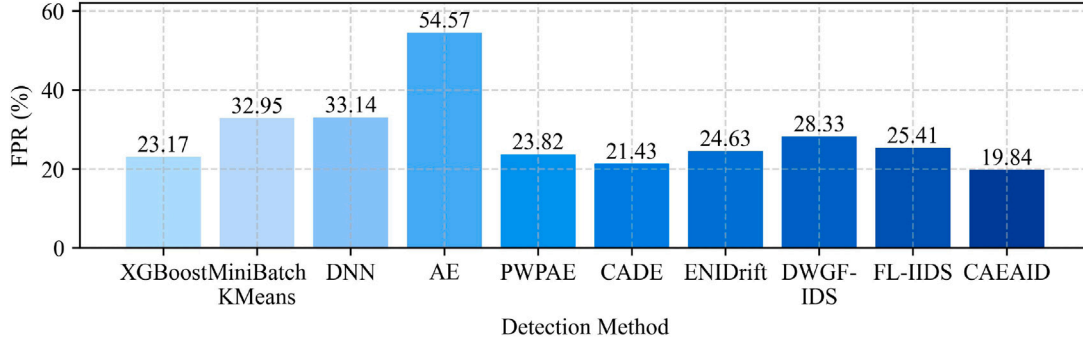


Fig. 9. Comparison of detection false positive rates of different methods on IoTID20.

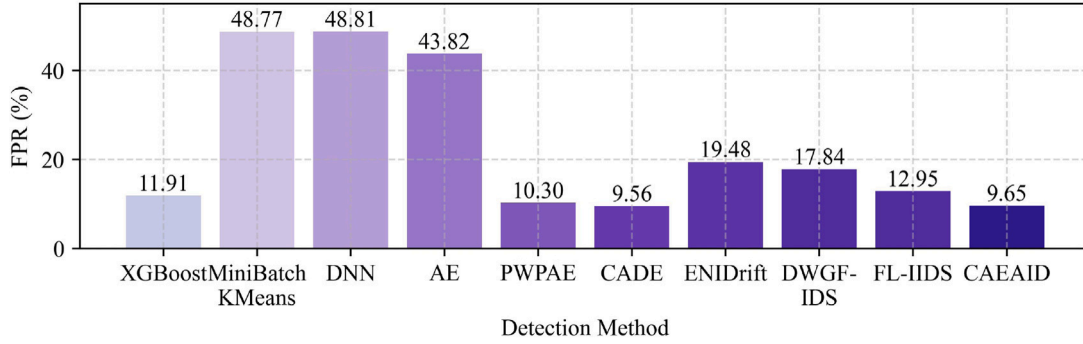


Fig. 10. Comparison of detection false positive rates of different methods on CICIDS2018.

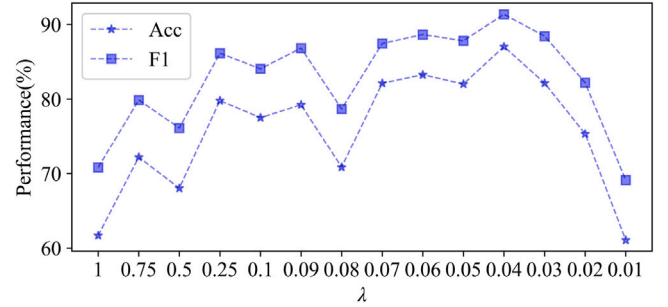
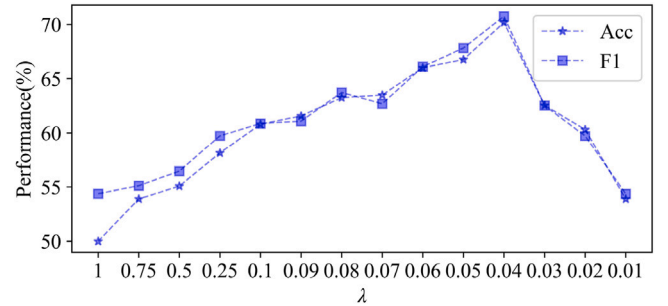
Table 5

Comparison of detection accuracy(%), precision(%), recall(%) and F1-score(%) of different methods on CICIDS2018.

Method	Accuracy	Precision	Recall	F1-score
XGBoost	72.80	95.84	46.26	62.41
MiniBatchKMeans	57.57	57.78	63.61	60.56
DNN	60.10	57.55	69.47	62.95
AE	56.32	56.31	56.47	56.39
PWPAE	88.85	89.52	87.99	88.75
CADE	89.68	90.29	88.92	89.60
ENIDrift	87.17	82.81	91.82	87.08
DWGF-IDS	84.72	73.02	87.29	85.10
FL-IIDS	83.90	86.18	80.76	83.38
CAEAID	91.22	90.52	92.09	91.30

3.37% in F1-score. Our CAEAID framework achieved an accuracy of 91.22%, precision of 90.52%, recall of 92.09%, F1-score of 91.30% when detecting samples with sparse distribution and drift behavior. Although the precision is slightly lower than that of XGBoost, the overall detection performance remains robust. CAEAID also realize the highest recall rate. In accuracy, CAEAID shows an improvement of 1.71% over the advanced concept drift detection method CADE. Additionally, the F1-score is improved by 1.90% compared to CADE.

As shown in Fig. 10, the false positive rate of CAEAID is 9.65%. It achieves a significant reduction in the false positive rate compared to both traditional machine learning and deep learning models. It achieves a significant reduction in the false positive rate compared to both traditional ML and DL models. Although slightly lower than CADE in this aspect, the difference is minimal. It indicates that the CAEAID framework demonstrates superior performance in detecting novel attack traffic compared to other state-of-the-art methods. The above results underscore the effectiveness of the CAEAID framework in addressing sparse data distribution and concept drift in IoT intrusion detection.

Fig. 11. Intrusion Detection Accuracy and F1-score of the Original Contrastive Autoencoder on IoTID20 at Different Values of λ .Fig. 12. Intrusion Detection Accuracy and F1-score of the Original Contrastive Autoencoder on CICIDS2018 at Different Values of λ .

4.2.2. Experiment on the setting of loss function parameter λ

An essential parameter in the loss function of the contrastive autoencoder is the weight λ of the contrastive loss. We evaluate the sensitivity of the performance of the contrastive autoencoder model

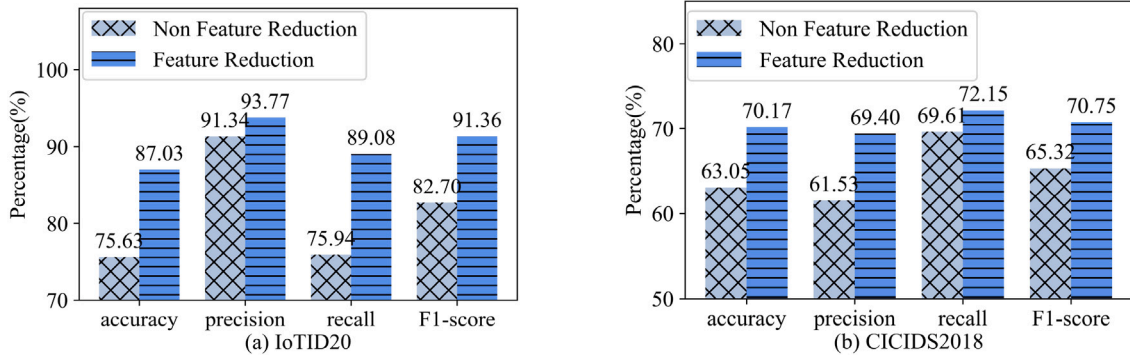


Fig. 13. Intrusion Detection Performance of Contrastive Autoencoders Trained on Traffic Data Before and After Feature Reduction.

to this parameter. Our experimental method involves setting up an ablation study, keeping other parameters constant while only varying λ . We set λ to values of 1, 0.75, 0.5, 0.25, 0.1, 0.09, 0.08, ..., and 0.01. For each value, we observe the detection accuracy and F1-score of the contrastive autoencoder after training solely on the original training data, as shown in Figs. 11–12.

According to Fig. 11, when $\lambda > 0.04$, as the value of λ decreases, the detection accuracy and F1-score of the contrastive autoencoder-based intrusion detection model on the IoTID20 dataset show fluctuations but exhibit an overall increasing trend. At $\lambda = 0.04$, the contrastive autoencoder achieves the highest detection accuracy and F1-score on the testing data from the IoTID20 dataset. When $\lambda < 0.04$, as λ continues to decrease, the accuracy and F1 score of the detection model drop sharply. Similarly, based on Fig. 12, when $\lambda > 0.04$, as the value of λ decreases, the detection accuracy and F1-score for new normal traffic and attack traffic in the CICIDS2018 dataset also increase continuously. At $\lambda = 0.04$, the detection accuracy and F1-score are at their highest. When $\lambda < 0.04$, as λ continues to decrease, the accuracy and F1-score of the detection model continuously decline. Therefore, we set $\lambda = 0.04$.

4.2.3. Validation of the effectiveness of feature reduction

To validate the effectiveness of feature reduction based on KDF and BP joint analysis, we trained the contrastive autoencoder-based intrusion detection model using both the original traffic feature data and the feature-reduced traffic feature data on IoTID20 and CICIDS2018. We used the trained models to detect the traffic data to be tested and compared the detection accuracy, precision, recall, and F1-score. The experimental results are shown in Fig. 13.

According to Fig. 13, it can be observed that training the contrastive autoencoder with the feature-reduced traffic training set results in higher detection accuracy, precision, recall, and F1-score. On the IoTID20 IoT dataset, the accuracy of the contrastive autoencoder improved by 15.08%, the precision increased by 2.66%, the recall rose by 17.30%, and the F1-score improved by 10.47% after feature reduction. Similarly, on the CICIDS2018 dataset, the contrastive autoencoder also significantly improved these four performance metrics after feature reduction. This demonstrates that after feature reduction, the CAEAID framework retains highly discriminative traffic statistical features for IoT intrusion detection while mitigating interference from less representative features during learning and detection processes. It validates the effectiveness of the feature reduction process.

4.2.4. Validation of the effectiveness of model updates

To validate the effectiveness of model updates, we compared the intrusion detection performance of the model under different update cycles. First, for CICIDS2018, we accumulated new traffic samples at 20,000, 40,000, 60,000, 80,000, 100,000, and the total amount of new traffic to be tested. The pseudo-labels and their corresponding traffic samples, determined by the anomaly detection based on improved extreme value theory and the contrastive autoencoder-based intrusion

detection model, were used to retrain the contrastive autoencoder along with the original traffic training samples. We then compared the intrusion detection performance under different conditions, with the results shown in Tables 6 and 7.

According to Table 6, during different update cycles of the model, the updated model shows significant improvements in detection accuracy, precision, recall, and F1-score for the traffic data and new attack traffic within that period compared to the non-updated model, along with a notable reduction in false positive rates. As shown in Table 7, with the continuous accumulation of samples, the number of traffic samples used for retraining increases, leading to a marked improvement in the detection performance of the updated model. It demonstrates that the CAEAID framework maintains strong generalization through incremental model updates and confirms its robust detection performance when addressing concept drift caused by traffic distribution shifts in IoT intrusion detection.

On the IoTID20 dataset, due to the smaller data volume, we only compared the intrusion detection performance before and after the model update, as shown in Table 8. The CAEAID framework achieved significant improvements in detection accuracy, precision, recall, and F1-score on the IoTID20 dataset, with increases of 8.57%, 5.18%, 5.38%, and 5.28%, respectively, along with a notable decrease in the false positive rate. The above findings validate the necessity and effectiveness of incremental learning and model updates in enabling the CAEAID framework to address IoT intrusion detection and combat concept drift.

4.2.5. Computational efficiency and memory usage analysis

We evaluated the CAEAID framework's computational efficiency and memory usage to assess its scalability. Computational efficiency was quantified by measuring runtime duration, while memory usage was tracked via system resource consumption. On the IoTID20 dataset, CAEAID achieved a total training and update time of 32.64s and detection time of 5.02s, with 367.8MB memory usage. For the larger CICIDS2018 dataset, it required 55.79s for training/updates and 31.16s for detection, consuming 978.5MB memory. The framework demonstrates acceptable computational efficiency and memory footprint, confirming its scalability for real-time IoT intrusion detection with balanced resource utilization.

5. Conclusion

To address the issue of rapid performance degradation in existing one-time machine learning models when faced with sparse distribution and concept drift in IoT intrusion detection traffic samples, we propose an incremental contrast learning-based intrusion detection framework for IoT networks, named CAEAID. The framework employs a contrastive autoencoder to obtain stable generalized feature representations of traffic data. An improved extreme value theory enhances attention to sparse discrete tail data in IoT network traffic, assisting

Table 6

Comparison of detection performance (%) of the model against the traffic data to be measured in the cycle after periodic model update on the CICIDS2018.

The number of samples accumulated	Model update or not	Accuracy	Precision	Recall	F1-score	FPR
20000	No	69.42	67.20	75.85	71.27	37.01
	Yes	75.41	72.96	80.73	76.65	29.91
40000	No	72.61	70.59	77.53	73.90	32.31
	Yes	86.23	87.32	84.77	86.03	12.30
60000	No	75.79	70.62	88.33	78.49	36.75
	Yes	90.17	84.71	98.05	90.89	17.10
80000	No	71.39	71.72	71.76	71.74	29.00
	Yes	86.58	82.45	92.95	87.38	19.79
100000	No	74.58	74.53	74.69	74.61	25.53
	Yes	87.58	85.38	90.69	87.96	15.53
244928	No	70.17	69.40	72.15	70.75	31.81
	Yes	91.22	90.52	92.09	91.30	9.65

Table 7

Comparison of detection performance(%) of the model against all traffic data to be measured after periodic model update on CICIDS2018.

Samples accumulated size	Acc	Pre	Rec	F1	FPR
0	70.17	69.40	72.15	70.75	31.81
20000	72.47	71.75	74.13	72.92	29.18
40000	73.44	73.48	74.03	73.75	27.16
60000	78.47	79.71	76.39	78.02	19.44
80000	81.40	80.83	82.32	81.57	19.52
100000	84.85	83.77	86.43	85.08	16.74
244928	91.22	90.52	92.09	91.30	9.65

Table 8

Comparison of detection performance(%) of the model against all traffic data to be measured after periodic model update on IoTID20 dataset.

Method	Acc	Pre	Rec	F1	FPR
Unincremental learning	80.16	89.15	84.53	86.78	34.47
Incremental learning	87.03	93.77	89.08	91.36	19.84

the contrastive autoencoder in determining pseudo-labels for partial testing data. The pseudo-labeled testing data enables training data augmentation and model incremental updating, improving robustness against concept drift in IoT intrusion detection. Extensive experiments on IoTID20 and CICIDS2018 datasets demonstrate that the CAEAIID framework achieves detection accuracies of 87.03% and 91.22%, and F1 scores of 91.36% and 91.30%, respectively, effectively addressing data sparsity and concept drift in IoT intrusion detection. Compared to state-of-the-art IoT intrusion detection models, CAEAIID improves detection accuracy by at least 1.15% and 2.89%, further validating its superiority in tackling the aforementioned challenges.

In future work, we will struggle the development of an interpretable, efficient, and dynamically adaptive intrusion detection framework. We will further explore methods for network traffic data analysis with reasonable resources in ultra-large-scale IoT network environments to effectively address the complex attack patterns encountered in real-world deployment scenarios.

CRedit authorship contribution statement

Zinuo Yin: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Hongchang Chen:** Writing – review & editing, Validation. **Hailong Ma:** Supervision, Resources, Funding acquisition. **Tao Hu:** Writing – review & editing, Visualization, Software. **Luxin Bai:** Project administration, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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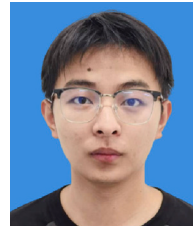
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